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Ambient solar thermal catalysis for polyolefin upcycling using copper encapsulated in silicon nanosheets and chloroaluminate ionic liquid

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Chuanwang Xing[1,8], Chengliang Mao[2,3,8], Shenghua Wang[1], Yuxuan Zhou[4], Lei Wu[2], Dake Zhang[1], Dingxuan Kang[1], Di Yang[3], Weiting Gong[3], Wendong Wei[3], Liang Wang[5], Chaoran Li[4], Geoffrey A. Ozin[2] ✉, Deren Yang[1,6] ✉ & Wei Sun[1,7] ✉

The accumulation of plastic waste has become a global issue. Socially and industrially viable, sustainable technical solutions are therefore required. Here we report a solar thermal catalytic system for polyolefins upcycling using copper nanoparticles encapsulated by stacked two-dimensional silicon. In a chloroaluminate ionic liquid solvent, unlike conventional thermal techniques, the upcycling can proceed under a mild temperature (55 °C) created photothermally under 4 sun irradiation. The polyethylene can be completely transformed into distinct and separable fractions of alkanes (C_3–C_7) and cyclic hydrocarbons (C_8–C_{26}) within hours, with a total yield of 91%. Mechanistic studies show a pathway involved two β-scissions of C–C bonds and a rapid cyclization. The approach offers versatility in the upcycling of various real-world polyolefin waste and features excellent feasibility in outdoor practices. The analyses of a conceptual upcycling facility using this technology showcase its appeal in both economic and eco-friendliness.

The worldwide plastics production has soared in the recent decades. It reached a staggering 390.7 million tonnes in 2021, with a substantial increase of 4% over the previous year[1–3]. Such gigantic plastic production causes severe greenhouse gas emissions throughout the entire life-cycle of the plastic[4]. Based on current growth trends, in 2050 15% of the global carbon budget will be taken up by the carbon emissions derived from plastics consumption[4]. Globally, only 9% of plastic waste is recycled—most of it is destined for landfills, incineration or leaked into the nature as waste, resulting in serious environmental and health issues[5]. It is therefore imperative to design approaches for plastic waste management.

Among the plastic recycling approaches, chemical upcycling is a promising technique that can transform waste into value-added products, allowing synchronous environmental remediation and chemical production[6]. However, its industrial practice is challenging because the economic value of the corresponding products cannot recover the costs of the upcycling process. In practice, depolymerization of polyolefins—which account for >50% of global plastic materials and waste—is constrained by the presence of extremely inert C–C bonds[7]. Thermocatalytic hydrogenolysis techniques for polyolefins have demonstrated some progress in recent years. However, the harsh operating conditions—including high temperatures (~300 °C), high hydrogen pressure, and the associated special reaction equipment and massive energy input—considerably increase the cost of polyolefin upcycling[8–10]. To tackle such problems, it has been recently been reported that polar environments can stabilize the ionic intermediates of plastic cracking, thereby lowering the reaction temperature[11–14]. This method has been applied in the recycling of polyethylene terephthalate

A full list of affiliations appears at the end of the paper. ✉e-mail: g.ozin@utoronto.ca; mseyang@zju.edu.cn; sunnyway423@zju.edu.cn

(PET) and polyethylene (PE). For instance, Zhang and colleagues employed a highly polar reaction environment using an ionic liquid solvent[13]. They coupled endothermic cracking of polymer C–C bonds with exothermic alkylation to overcome the reaction barrier. With isopentane (iC_5) as the co-reactant, low-density polyethylene (LDPE) was completely converted into isoalkanes under a low temperature of 70 °C. However, this strategy consumes a stoichiometrically equivalent number of iC_5 chemical to recycle cheap plastic waste, which may restrict its economy of scale.

Solar-driven catalysis has drawn considerable interest as another green and energy-saving method to upcycle plastic waste. Some well-explored photocatalysts for traditional aqueous photocatalysis, including Nb_2O_5 (ref. 15), VPOM/g-C_3N_4 (ref. 16) and Zr-doped $CoFe_2O_4$ (ref. 17), have been adopted to depolymerize polyolefins via excited-state carrier-driven redox reactions under light. Unfortunately, although these photochemical recycling approaches can proceed under mild conditions, their extremely low yield rates (typically at the μmol $g_{cat.}^{-1}$ h^{-1} level) prevent practical application. Meanwhile, as an emerging plastic upcycling approach, photothermal catalysis stands out with clean energy input and excellent conversion efficiency[4,18–20]. Very recently, Zhang and co-workers reported a photothermal system for the hydrogenolysis of polyolefin, reaching almost complete transformation of the plastic waste in a short period (3 h)[18]. Nevertheless, a high light intensity of tens of suns is required to reach the required operating temperature, resulting in a small irradiated area due to the employment of solar concentrator. Thus, opportunities remain to develop advanced photothermal catalysts and systems for promoting the feasibility and scalability of green, mild and efficient upcycling of polyolefins.

Here we design a low-temperature and co-reactant-free solar thermal system to catalytically upcycle polyolefins into high-value hydrocarbons. Cu/2D Si heterostructure catalysts were prepared by encapsulating copper nanoparticles (NPs) in stacked two-dimensional silicon (2D Si). The chloroaluminate ionic liquid, $[C_4Py]Cl\text{-}AlCl_3$, was diluted in chloroform as the solvent to establish a highly polar reaction environment. The two components of Cu/2D Si can synergistically capture the whole solar spectrum, stimulating local photothermal heating to drive the upcycling. Notably, LDPE plastics can be completely converted under a relatively low temperature of 55 °C, photothermally raised by a moderate light intensity of 4 sun (400 mW cm^{-2}) within 6 h. The products with a distinct distribution consist of valuable alkanes (C_3–C_7) and cyclic hydrocarbons (C_8–C_{26}), and the total yield is 91%. Mechanistic studies revealed a selective plastic cracking pathway, involving two β-scissions of C–C bonds and a rapid intramolecular cyclization. The Cu NPs also act as active sites for the dehydrogenation process, boosting conversion efficiency. Furthermore, the solar thermal catalysis offers versatility in the upcycling of various real-world polyolefin waste and features excellent feasibility in an outdoor demonstration. Based on techno-economic analysis and life-cycle assessment (LCA), converting 1 tonne of LDPE waste can be economic viable in certain regions and reduces greenhouse gas emissions by 30% compared with conventional fossil-based routes.

Results

Synthesis and characterization of catalysts

Our strategy used Cu/two-dimensional 2D Si as the catalyst, combining the outstanding dehydrogenation activity of metallic Cu NPs and excellent solar light absorption properties of silicon nanosheets[21–23]. The 2D Si was exfoliated from bulk calcium silicide. The 2D Si seems to be a stacked layer structure, which consists of hydride-terminated silicon nanosheets (Supplementary Fig. 1). The Cu NPs were then grown on the 2D Si surface through a facile wet-impregnation method (Fig. 1a). During the process, the hydrides on the silicon nanosheets surface in situ-reduced Cu(II) to Cu(0) (ref. 24), resulting in Cu NPs embedded in the stacked 2D Si structure. The variation of surface silicon hydrides and the formation of metallic Cu were verified by X-ray diffraction (XRD), infrared spectroscopy and high-resolution X-ray photoelectron spectroscopy (Supplementary Figs. 2 and 3). The morphological information of Cu/2D Si was collected by transmission electron microscopy (Fig. 1b–d). Although most of the Cu NPs were encapsulated inside of the stacked 2D Si, making observation via transmission electron microscopy (TEM) difficult (Fig. 1b and Supplementary Fig. 4a,b), we can identify the brighter spots of Cu NPs inside of the 2D Si from the high-angle annular dark field TEM (HAADF-TEM) images; these Cu NPs have an average diameter of 34.3 ± 7.5 nm (Fig. 1c and Supplementary Figs. 4 and 5). The lattice fringes shown in Fig. 1c were determined to be 0.22 nm, matching the (111) planes of metallic Cu (JCPDS no. 04-0836). Moreover, the energy-dispersive X-ray spectroscopy mapping (Fig. 1d) shows that the Si, O and Cu elements are distributed on their components, whereas the oxygen element originates from the inevitable oxidation during the preparation of 2D Si. The confinement of Cu NPs within 2D Si can lead to the strong metal–support interaction, and thus boost the catalytic stability (refer to the more detailed discussion below)[21,25].

The X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) were tested to investigate the chemical environment of Cu in Cu/2D Si with atomic-level insight. The Cu K edge XANES spectra of Cu/2D Si and reference samples, and their corresponding first derivatives, are presented in Fig. 1e. The Cu–K adsorption energy of Cu/2D Si resembles that of Cu foil (8,980.3 eV), which further substantiates the idea that the Cu NPs are mainly in the metallic state[26]. The k^3-weighted and not phase-corrected Cu K EXAFS data in R-space and corresponding fitting results are shown in Fig. 1f, Supplementary Fig. 6 and Supplementary Table 1. The Cu/2D Si shows obvious Cu–Cu scattering paths at 2.57 Å (posterior to phase correction), originating from the metallic copper[27]. Besides, the sample also exhibits first-shell scattering at 1.93 Å, which is proximate to the cases for CuO (1.94 Å)[28]. This result illustrates the presence of Cu–O species in Cu/2D Si sample, indicating a strong interaction between Cu and partially oxidized 2D Si (ref. 29). The wavelet transforms conducted upon the EXAFS spectra further verified the phenomenon. Cu/2D Si catalyst displays an intensity maximum located at $k = 7.5$ $Å^{-1}$ and $R = 2.2$ Å, assigned to the Cu–Cu scattering path, and another area with minor intensity centred at $k = 7.9$ $Å^{-1}$ and $R = 1.5$ Å, assigned to the Cu–O scattering path[30].

Solar thermal plastic upcycling scheme

We aimed to develop a green and mild solar thermal approach to upcycling polyolefin waste. As the scission of polymer C–C bond usually undergoes carbenium ion transition states, we adopted an ionic liquid to establish a highly polar reaction environment for stabilizing the intermediates as reported[13,31]. The solar thermal catalytic system is shown in Fig. 2a. The experiments were performed in a quartz tube reactor using chloroaluminate ionic liquid diluted in chloroform as the solvent, whereas the chloroform was chosen due to its inertness in the system, good solubility with $[C_4Py]Cl\text{-}AlCl_3$ and its higher boiling point than dichloromethane (see Supplementary Fig. 7 and Supplementary Note 1 for details). Typically, a mixture of 10 mg of Cu/2D Si as the solar thermal catalyst, 200 mg of LDPE (weight-averaged molecular weight (M_w) of 157,570 g mol^{-1}, number-averaged molecular weight (M_n) of 20,751 g mol^{-1}) as the substrate and 10 μl of *tert*-butyl chloride (TBC) as the initiator were dispersed in the solvent. The M_w of LDPE in this work exceeds that of most studies[9,13,18], making its conversion more challenging. The chloroaluminate ionic liquid may undergo hydrolysis when a considerable amount of water is present, which can lead to deactivation (Supplementary Fig. 8); therefore, the conversion process was conducted under an inert atmosphere. The reactor was under simulated sunlight illumination from a xenon lamp, stimulating the photothermal effect of Cu/2D Si. Coupling the interband transition of 2D Si and the localized surface plasmonic resonance effect of

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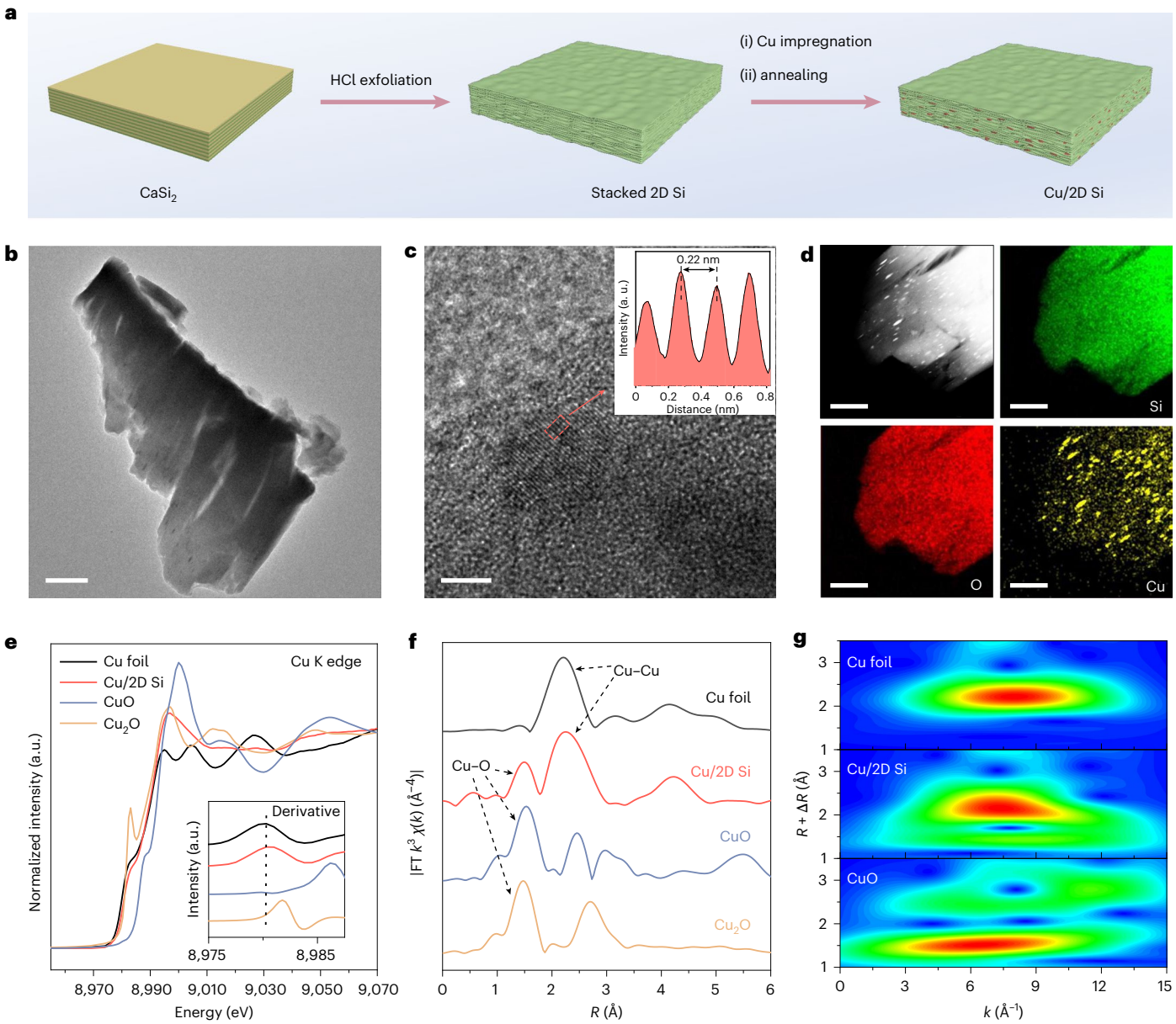

Fig. 1 | Schematic preparation and characterization of Cu/2D Si catalyst. **a**, Illustration of preparation of 2D Si and subsequent encapsulation of Cu NPs. **b**, TEM image of Cu/2D Si. Scale bar, 500 nm. **c**, High-resolution TEM of Cu/2D Si. Scale bar, 3 nm. Inset: the intensity profile of the area indicated in **c**. a.u., arbitrary units. **d**, HAADF-TEM image and energy-dispersive X-ray spectroscopy mappings of Cu/2D Si. Scale bars, 300 nm. **e**, Cu K-edge XANES spectra. Inset: first derivatives. **f**, Fourier-transformed (FT) k^3-weighted EXAFS spectra of Cu/2D Si and reference samples. **g**, Wavelet-transform contour plots of Cu foil, Cu/2D Si and CuO samples.

Cu NPs, the Cu/2D Si hybrid can efficiently harness the entire solar spectrum (Supplementary Fig. 9)[32]. The photoinduced carriers then generate local heat through the non-radiative relaxation to increase environmental temperature. When the light intensity reached 4 sun (400 mW cm^{-2}), the solution temperature raised to 55 °C within 10 min (Fig. 2d and Supplementary Fig. 10). The further increase of light intensity did not generate an obvious increase of the system temperature, which was probably due to severe heat loss caused by the volatilization of chloroform at temperatures near its boiling point. Therefore, the light intensity in the reaction was determined to be 4 sun for the following experiments. It should be noted that previously reported photothermal systems usually feature highly focused light with the high intensities of tens of suns. By contrast, our much lower light intensity allows a multiplied light incident area, which is particularly attractive for larger-scale solar-driven practices.

By this solar thermal catalytic approach using Cu/2D Si catalyst, LDPE can be completely converted within 6 h at a remarkably low temperature (55 °C, Fig. 2b). No gaseous or liquid products were detected in the absence of LDPE, verifying that the observed products originated from plastic upcycling. There is no local photothermal heating effect without light or Cu/2D Si catalyst, resulting in negligible LDPE conversion in the system. The ionic liquid is also vital for mild plastic deconstruction, as it stabilizes the ionic intermediates during C–C bond cracking, thereby lowering the reaction barrier. Control experiments showed that the system exhibited minimal LDPE conversion in the absence of chloroaluminate ionic liquid. The initiator, TBC, provides the initial concentration of carbo-cations via chloride abstraction with ionic liquid and boost the conversion rate (Supplementary Fig. 11). During the reaction, the gaseous products were evolved from the liquid mixture, which accumulated in the headspace and were analysed using

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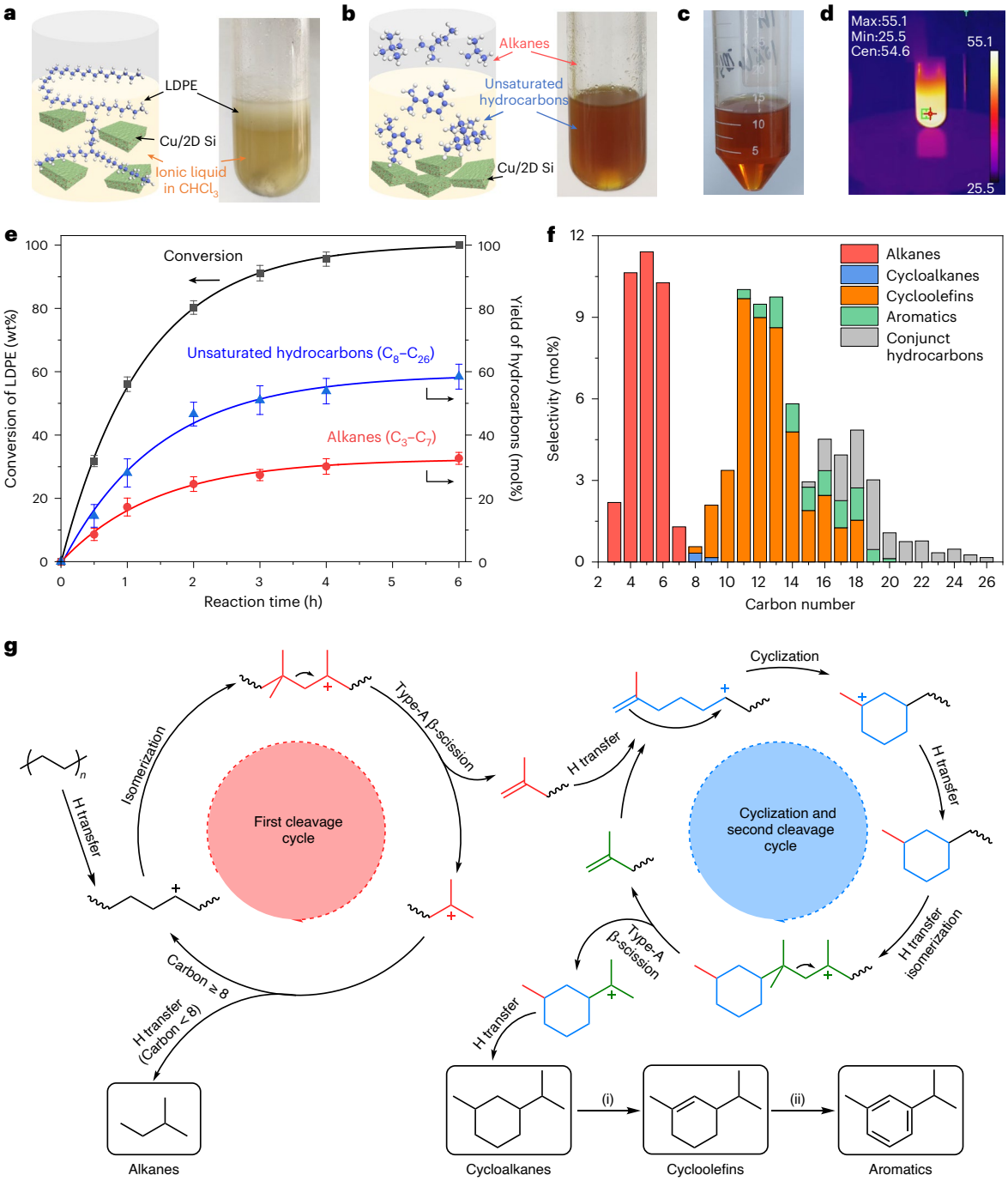

Fig. 2 | Performance and proposed reaction pathways for solar thermal catalytic plastic upcycling. a,**b**, Schematic illustration and snapshots of the photothermal LDPE upcycling system before (**a**) and after (**b**) reaction. Reaction conditions, 200 mg of LDPE powder; 10 mg of Cu/2D Si catalyst; diluted ionic liquid solvent (6 mmol of $[C_4Py]Cl$, 12 mmol of $AlCl_3$, and 3 ml of chloroform); 10 µl of TBC initiator; and 4 sun simulated solar irradiation for 6 h. **c**, Snapshot of liquid products after extraction by chloroform. **d**, Infrared thermography image of the photothermal upcycling system. Colour bar indicates temperature in °C. Temperature at the center (Cen) indicated by the green cross. **e**, Time-resolved conversion and yield profile (black squares, LDPE conversion; red circles, alkanes yields; blue triangles, unsaturated hydrocarbons yields). The curves represent the optimal fit to the data. All measurements were repeated five times, and the results are presented as mean values with s.d. error bars. **f**, Distribution of the hydrocarbon products formed over Cu/2D Si catalyst. The results are presented as mean values of five independent measurements. **g**, Proposed reaction pathways for the selective cleavage of LDPE via type-A β-scission of the C–C bond. The cyclohydrocarbons are formed by a rapid intramolecular cyclization and break away from the polymer chain via two type-A β-scissions. Red, blue and green structures correspond to parts in the first type-A β-scissions, cyclization and the second type-A β-scissions, respectively. Cycloalkanes transform route: (i) dehydrogenation; (ii) dehydroaromatization.

a gas chromatograph (Supplementary Fig. 12). Notably, the colour of the liquid mixture turned dark brown as the reaction proceeded, indicating the formation of unsaturated cyclic hydrocarbons[33]. The liquid products were extracted with chloroform, and the distribution and yield were determined by a combination of gas chromatography, gas chromatography–mass spectrometry (GC–MS) and two-dimensional GC–MS (GC×GC–MS) measurements (Fig. 2c, Supplementary Figs. 13 and 14, and Supplementary Note 2). Concretely, alkanes range from C_3

Table 1 | Summary of performance of catalytic plastic upcycling under different reaction conditions

Catalyst	Mechanism[a]	LDPE conversion (%)	Product yield (%)	Product selectivity (%)					Average carbon number of product
				Alkanes[b]	Cycloalkanes[b]	Cycloolefins[b]	Aromatics[b]	Conjunct hydrocarbons[b]	
Cu/2D Si	Photothermal	100	91.2	35.8	0.5	44.7	7.5	11.4	10.8
2D Si	Photothermal	58.1	54.0	33.6	0.7	53.2	1.8	10.7	11.0
Cu/SiO_2	Photothermal	38.3	37.2	33.8	0.7	35.5	12.9	17.1	11.5
Cu/2D Si	Thermal	39.4	36.4	31.1	0.5	49.8	4.7	13.9	11.5
Cu/2D Si	Thermal+UV	56.2	53.2	46.7	7.7	37.5	3.0	5.1	9.7
Cu/2D Si	Sunlight	100	85.8	37.2	0.3	45.9	7.4	9.2	10.9

[a]Photothermal catalytic system is under 4 sun simulated solar irradiation. Thermal catalytic system is maintained at 55 °C by heating. The intensity of UV light is 251 mW cm^{-2}. The intensity of incident sunlight was 4 to 5 sun. [b]Products are classified on the basis of the degree of unsaturation (0, alkanes; 1, cycloalkanes; 2 and 3, cycloolefins; 4, aromatics; >4, conjunct hydrocarbons), calculated from their MS signals.

to C_7 with a total yield of 35.8%, whereas cyclic hydrocarbons spread from C_8 to C_{26} with a total yield of 55.4% (Fig. 2e). Low-value methane and ethane were not detected in the products. The hydrogen-to-carbon ratio of alkanes (with the formula C_nH_{2n+2}) is larger than that of the substrate polyethylene (with the formula C_nH_{2n}). Hence, highly unsaturated cyclic products were generated to maintain the stoichiometric equilibrium in the upcycling process. As shown in Fig. 2f and Table 1, depending on the formula calculated from MS signals, the products were classified into four categories: cycloalkanes, cycloolefins, aromatics and conjunct hydrocarbons (that is, compounds with more than four degrees of unsaturation, presenting a structure integrated from the first four categories). The product distribution was measured five times, and the corresponding data are presented in Supplementary Fig. 15.

Notably, unlike most past reports, the alkanes and unsaturated hydrocarbons in our work are distributed in distinct carbon number ranges. The alkanes distribution ends at *n*-heptane, which exhibits the highest boiling point of 98 °C among the alkane products, whereas the distribution of unsaturated hydrocarbons starts from dimethylcyclohexanes, which exhibit the lowest boiling point of about 121 °C. Furthermore, the plastics upcycling process displayed particularly low selectivity for C_7 and C_8 products. Thus, the saturated and unsaturated hydrocarbons can be feasibly separated according to their large difference in boiling point. To validate this, a proof-of-concept experiment was conducted by distiling the liquid-phase products. By comparing the hydrocarbon distribution before (Supplementary Fig. 16a) and after (Supplementary Fig. 16b) distillation, we observed that most linear hydrocarbons ranging from C_1 to C_7 had been removed. Among the remaining liquid-phase products, selectivity for cyclic hydrocarbons exceeded 97 mol%. These findings suggest that our system has strong potential for facile separation of valuable cyclic hydrocarbons. Besides, in the field of polyolefin upcycling, our solar thermal catalytic system demonstrates a superior production rate even compared with the reported thermal systems with co-reactants at high temperatures (Supplementary Table 2).

Based on the product distribution, detailed plastic upcycling reaction pathways are proposed in Fig. 2g. The radical reaction mechanism was eliminated via injecting radical scavengers into the system (Supplementary Fig. 17). The deficiency of methane and ethane indicated that upcycling occurs overwhelmingly via inner β-scission rather than terminal C–C cleavage (Supplementary Fig. 18)[13]. In principle, β-scission can be classified into five types on the basis of the branching degree of starting and ending carbenium ions (Supplementary Fig. 19)[34]. As the relative scission rate of carbenium increases with its branching degree, the type-A β-scission mechanism dominates the C–C bond cleavage of LDPE[35]. In the initial stage, the polyolefin chain transforms into carbenium ions via hydride transfer[36]. Due to the high formation energy of primary carbenium, the polymeric skeleton will preferentially isomerize via consecutive hydride- and methyl-shifts to more stable tertiary carbenium ions. The tertiary carbenium then undergoes type-A β-scission, producing a shorter tertiary carbenium ion and an olefin. As the carbo-cations for type-A β-scission require at least eight carbon atoms (Supplementary Fig. 18), the formed tertiary carbenium with fewer than eight carbon atoms will undergo hydride transfer to produce alkanes, whereas other longer tertiary carbenium can be further cracked in the cleavage cycle (Fig. 2g). For the olefins, carbo-cations can be generated on their polymer chain via hydride transfer. When the carbo-cation is positioned two to four carbon atoms away from an adjacent C=C bond, a rapid cyclization occurs, producing a cyclic structure on the chain[31]. Then, on the long-chain side of the cyclic structure, another type-A β-scission occurs following the hydride transfers and isomerization, leading to the generation of small-molecule cycloalkanes (Fig. 2g). The cycloalkanes can undergo dehydrogenation to generate cycloolefins and aromatics, and further proceed cyclization and isomerization to form polycyclic conjunct hydrocarbons. The detected MS signals of products in our reaction agree well with this pathway (Supplementary Fig. 20).

In brief, the cyclohydrocarbon breaks away from the polymer chain following β-scission at each end, followed by rapid cyclization. Interestingly, limited by the branching degree of type-A β-scission product, the cyclohydrocarbons obtained via the two type-A β-scissions have at least ten carbon atoms (the structure of the smallest cycloalkanes by this pathway is exhibited in Fig. 2g). Similarly, generation of C_8 and C_9 cyclohydrocarbons have to involve type-B β-scission, whereas the C_7 cyclohydrocarbons is absent owing to the unfavourable direct β-scission of monobranched carbo-cations (Supplementary Figs. 21 and 22). The carbo-cations with seven carbon atoms can further crack via type-B β-scission, which elucidates the particularly low selectivity of C_7 alkanes. Our production distribution forcefully supports the reaction pathway.

Functions of Cu NPs and 2D Si in LDPE upcycling

On the basis of the reaction pathways, we further explored the specific function of each active component. To reveal the function of Cu NPs on plastic upcycling, we prepared a control Cu/SiO_2 material by supporting Cu NPs on inert SiO_2. The TEM images show that the Cu NPs in Cu/SiO_2 exhibit the same crystal structure and particle size as those in Cu/2D Si (Supplementary Fig. 23). We investigated their catalytic properties under simulated sunlight for 6 h. Pure 2D Si and Cu/SiO_2 exhibited a lower LDPE upcycling performance (conversion: 58.1% for 2D Si, 38.3% for Cu/SiO_2; yield: 54.0% for 2D Si, 37.2% for Cu/SiO_2) than Cu/2D Si (Fig. 3a). The solid residues extracted after 3 h reaction were investigated by gel permeation chromatography (GPC; Fig. 3b).

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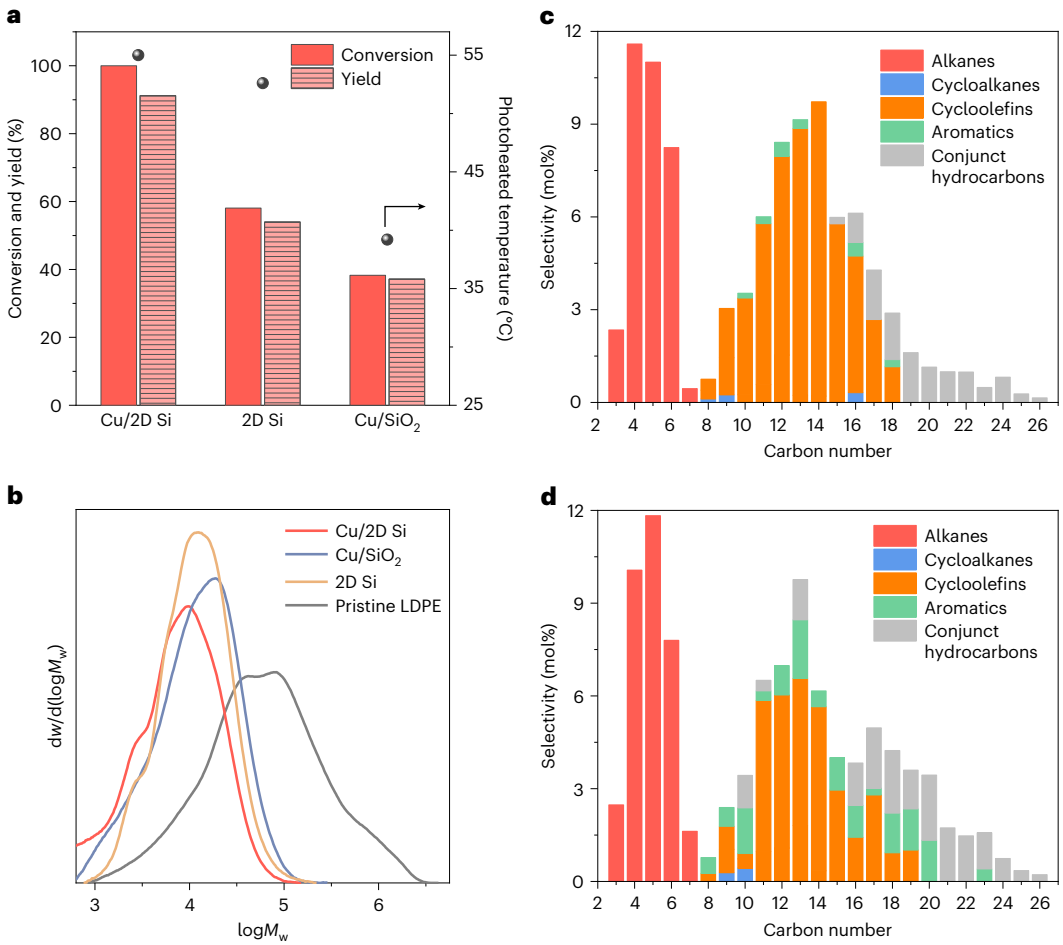

Fig. 3 | Function of Cu NPs and 2D Si in solar thermal catalytic plastic upcycling. a, Conversion of LDPE and yield of hydrocarbons over Cu/2D Si, 2D Si, Cu/SiO_2 catalysts under 4 sun simulated solar light after 6 h reaction. **b**, GPC analysis of the molecular weight of pristine LDPE and LDPE residue after 3 h reaction over Cu/2D Si, 2D Si, Cu/SiO_2 catalysts under 4 sun simulated solar light. **c**,**d**, Distribution of the hydrocarbon products formed over 2D Si (**c**) and Cu/SiO_2 (**d**) catalysts under 4 sun simulated solar light after 6 h.

The M_w of polyethylene sharply dropped from 157,570 g mol^{-1} (for pristine polyethylene) to 9,242 g mol^{-1}, 15,141 g mol^{-1} and 17,257 g mol^{-1} for Cu/2D Si, 2D Si and Cu/SiO_2, respectively. This result is consistent with the catalytic performance. The coupling solar capture of Cu and Si could produce a stronger local heating effect than individual components, which is supported by the fact that Cu/2D Si hybrid exhibited the highest apparent temperature of all of the samples (Supplementary Fig. 24). Combining the UV–vis–infrared diffuse reflectance spectra and photoheating results, the 2D Si was identified as the primary absorber in the hybrid that converts sunlight into heat.

In addition to photoheating of 2D Si, the Cu NPs also affect the catalytic properties. The absence of Cu species induced a substantial decline in the hydrocarbon yield for the photothermal catalytic plastic upcycling, even when excluding the influence of temperature (Supplementary Fig. 25). The role of Cu was further analysed from the product distribution. As shown in Fig. 3c,d and Table 1, the cycloolefins proportion decreased from 2D Si, to Cu/2D Si, to Cu/SiO_2, whereas the aromatics proportion increased from 2D Si, to Cu/2D Si, to Cu/SiO_2. We therefore speculate that the loading of Cu NPs promoted the dehydrogenation of cycloalkanes (Fig. 2g), which is in line with what has been reported previously[37,38]. We designed a control experiment for the hydride transfer between 2-methylbutane (iC_5H_{12}) and deuterated hexane (C_6D_{14}), aimed at directly demonstrating the activity of Cu species. On Cu/2D Si, substantial quantities of $iC_5H_{12-x}D_x$ and $C_6D_{14-y}H_y$ were observed (Supplementary Fig. 26), whereas pure 2D Si shows little activity, suggesting that the metallic NPs facilitate the hydride abstraction and transfer of hydrocarbons. Considering the ubiquity of H species migration during depolymerization of polyolefins, the loading of Cu NPs on 2D Si impels the proceeding of overall reaction, thereby notably boost the catalytic performance. Thus, the upcycling reactions can take place at the polymer–catalyst interface in the heterogeneous system. Besides, the reactions can also occur at the polymer–solvent interface, as evidenced by some LDPE conversion in the absence of solid catalyst.

It is also found that the constituent of Cu/2D Si hybrids plays a crucial role in ruling their catalytic activity. As shown in Supplementary Fig. 27 and Supplementary Table 3, with increasing the mass ratio of Cu, the solar thermal product yield increased first and then decreased subsequently, and the sample with a Cu mass ratio of 3.5% determined with inductively coupled plasma mass spectrometry (that is, the sample discussed above prepared with the feeding metal mass ratio of 6%) presents the optimal performance. The result can be explained by the fact that an increased amount of Cu could help to facilitate the hydrogen transfer during the plastic upcycling; however, over time Cu NPs will inevitably accumulate on the surface of 2D Si and suppress the light absorption of silicon.

Function of light in LDPE upcycling

Having established the mechanism of Cu/2D Si catalyst, we further studied the specific influence of light in the solar thermal LDPE upcycling

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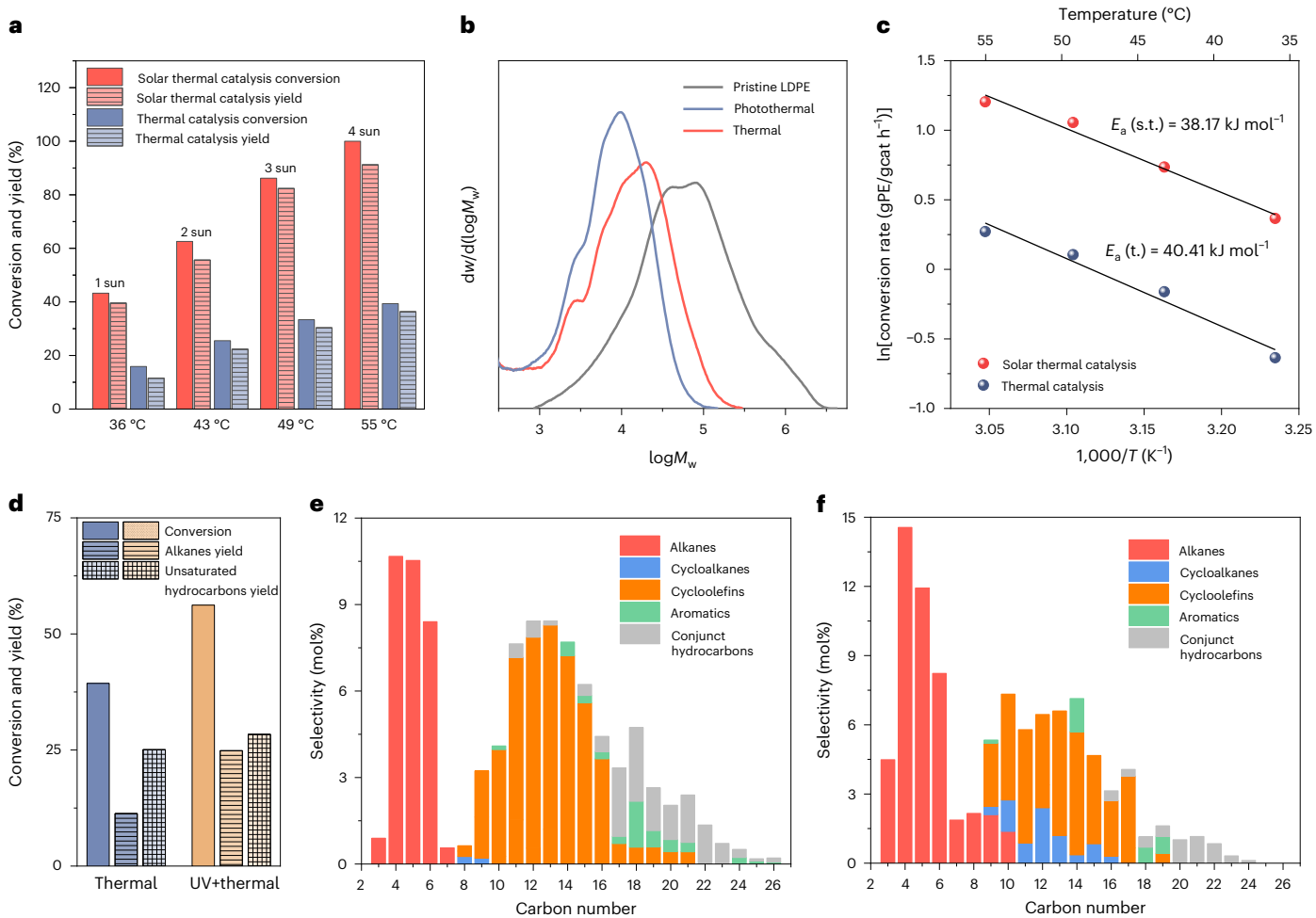

Fig. 4 | Function of light in solar thermal catalytic plastic upcycling. **a**, Impact of light intensity/temperature on the conversion of LDPE and yield of hydrocarbons under solar thermal and thermal catalysis. **b**, GPC analysis of the molecular weight of pristine LDPE and LDPE residue after 3 h reaction in solar thermal and thermal catalysis. **c**, Kinetics analysis reveals that the LDPE conversion rate over 6 h has a logarithmic dependence on the inverse temperature. E_a (s.t.), apparent activation barrier for solar thermal catalysis; E_a (t.), apparent activation barrier for thermal catalysis. **d**, Conversion of LDPE and yields of alkanes and unsaturated hydrocarbons under thermal and thermal plus UV catalysis. **e,f**, Distribution of the hydrocarbon products formed over Cu/2D Si catalyst under thermal (**e**) and thermal plus UV (**f**) catalysis.

system. Compared with external heating in traditional thermal catalytic systems, solar thermal catalysis enables local catalyst heating under clean solar radiation, thus minimizing energy consumption. As mentioned above, our Cu/2D Si can capture the full sunlight spectrum and locally generate heat by various non-radiative relaxation. The local heating effect of the Cu/2D Si can be demonstrated on the basis of heat transfer stimulation (Supplementary Fig. 28) and catalytic tests. As shown in Fig. 4a, the solar thermal catalytic upcycling of LDPE presents greatly enhanced performance compared with that under thermal catalysis at each temperature studied. The solid residues extracted after 3 h reactions were investigated by GPC (Fig. 4b). Consistent with the catalytic performance, the solid residue of solar thermal upcycling showed lower M_w than that of thermal upcycling.

A potential argument against this ambient solar thermal catalytic strategy is that the system may be alternatively driven by thermal catalysis powered by renewable electricity. We therefore compared the energy consumption of a direct solar thermal upcycling system under 1 sun and a thermal upcycling system at 55 °C since they feature similar catalytic performance and same treatment procedures. The thermal upcycling system is assumed to be heated by a 95% thermal efficiency heater powered by a 20% photoelectric conversion efficiency solar cell. As discussed in Supplementary Fig. 29 and Supplementary Note 3, estimations show that the direct solar thermal upcycling systems only require about one-quarter of the solar power (23.76 kJ) of thermal upcycling systems (91.61 kJ) to achieve a similar conversion and yield, signifying its superiority.

The performance enhancement may be contributed to the localized heating effect and/or the photochemistry effect. The kinetics analysis was thus performed to distinguish the mechanisms via the logarithmic dependence of the LDPE conversion rate over 6 h on the inverse temperature (Fig. 4c). However, solar thermal catalysis only exhibits a slightly lower apparent activation barrier than thermal catalysis, which cannot convincingly infer the underlying mechanism of LDPE depolymerization. Besides, the product distributions of thermal and solar thermal upcycling do not exhibit apparent variance (Fig. 4e and Table 1).

This phenomenon may be caused by the interference factor of UV light acting directly on the plastic. It is reported that UV photons can destroy the crystalline domains and break the C–C bonds (bond energy ≈ 3.44 eV) in LDPE[18]. We further irradiate a strong UV light (251 mW cm^{-2}) on the thermal upcycling system. It was found that the introduction of UV light enhanced the LDPE conversion and hydrocarbon yield (Fig. 4d). Besides, the incident UV light has a substantial impact on the product distribution. The light alkanes proportion prominently increased under the strong UV light (Fig. 4f), resulting in the decrease of the average carbon number of product from 11.5 for pure

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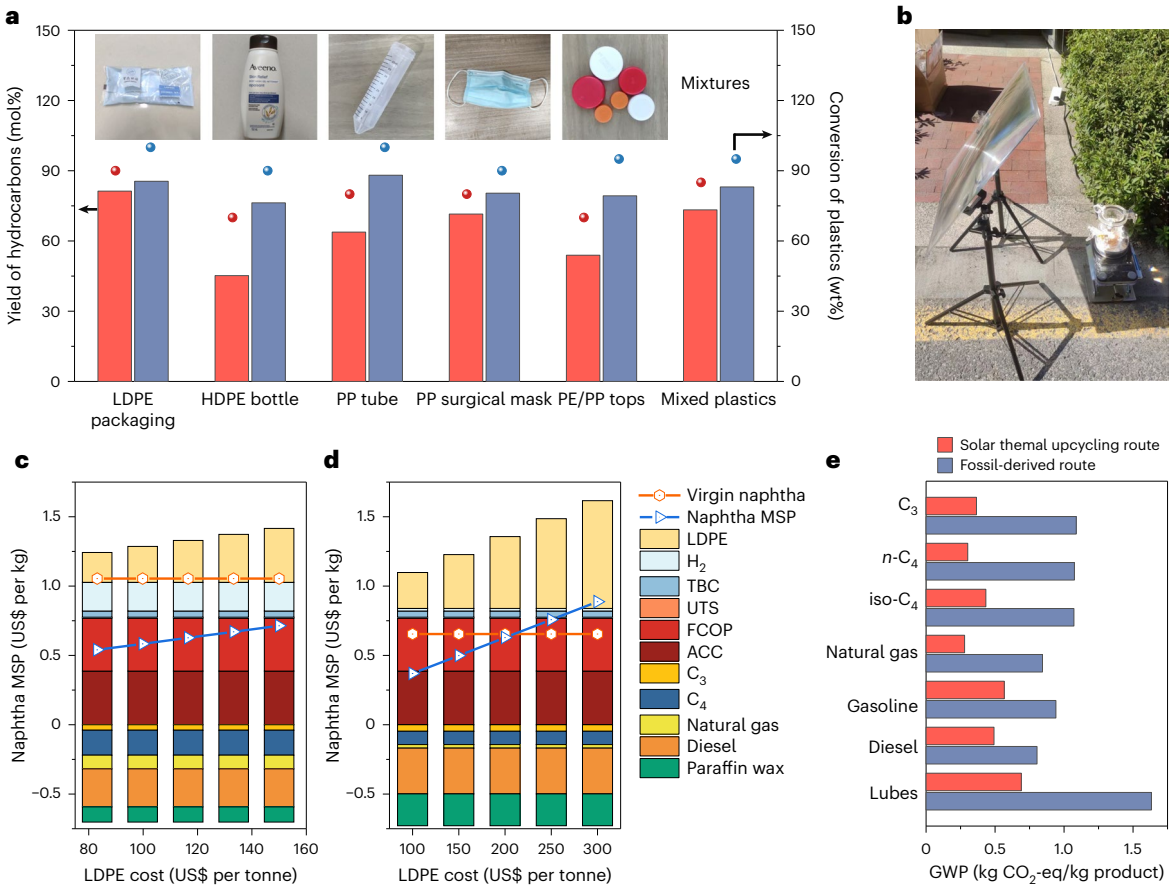

Fig. 5 | Robustness and demonstrations of solar thermal catalytic plastic upcycling system. a, Conversion of various real-world post-consumer plastics (bars) and yield of hydrocarbons (dots) over Cu/2D Si under 4 sun simulated solar. The red data were obtained from unpurified post-consumer plastics, whereas the blue data were obtained from plastics purified by dissolution–reprecipitation. **b**, Set-up of the outdoor experiments including a Fresnel lens to concentrate sunlight, a 300 ml reactor with a quartz window, a 5 g LDPE bag, 50 mg of Cu/2D Si catalyst, 30 mmol of $[C_4Py]Cl$, 60 mmol of $AlCl_3$, 15 ml of chloroform and 50 µl of TBC initiator. The incident intensity of the real sunlight was 4 to 5 sun. Reaction time = 6 h. **c**,**d**, Operational costs and product revenues breakdown for the solar thermal upcycling of LDPE waste in China (**c**) and in the United States (**d**). Orange hexagons, the virgin petroleum-derived naphtha price; blue triangles, naphtha MSP; bars above 0, operational costs per kilogram of naphtha produced; bars below 0, revenues per kilogram of naphtha produced. UTS, total utilities; FCOP, fixed costs of production; ACC, total annual capital charge. **e**, GWP of solar thermal LDPE upcycling compared with fossil-derived route for individual products.

thermal catalysis to 9.7. This is because the UV-initiated cross-linking reaction in polyethylene will generate more branched polymer chains. As the carbo-cations are formed preferentially on the tertiary carbon atoms, the β-scission of C–C bond in plastics is facilitated under UV light, causing more light alkanes in the product.

To investigate the pure photothermal effect of Cu/2D Si catalyst without the UV contribution, we explored LDPE upcycling under illumination with a an ultraviolet light filter applied (>420 nm allowed). The absorption edge of 2D Si and the absorption peak of localized surface plasmonic resonance effect of Cu NPs are both located in the visible region[39–41]. The photothermal upcycling system under visible light shows superior performance than that of thermal catalysis (Supplementary Fig. 30a). Interestingly, kinetics analysis signified a virtually consistent apparent activation barrier between the two systems (Supplementary Fig. 30b). Besides, the product distributions are also similar (Supplementary Fig. 30c). We therefore conclude that the dominant driving force for the boosted performance is the localized heating effect rather than the photochemical effect. Furthermore, the energy band diagram implies that the photoexcitation of 2D Si cannot produce electrons/holes with strong positive/negative potential to drive the redox of inert polymer C–C bond (Supplementary Fig. 31)[15]. Considering the very low proportion of UV light (~4%) in the solar spectrum, the solar thermal catalytic system can only receive few UV photons. This is why the solar thermal catalysis for LDPE upcycling does not exhibit a substantially different activation barrier and product distribution from those of thermal catalysis.

Robustness and application of solar thermal catalytic plastic upcycling system

To demonstrate the potential of practical applications of our plastic upcycling system, we first explored the robustness of the system. We performed repeated upcycling reaction to test the system's stability. In each repeated run, 200 mg of LDPE powder was refilled into the solar thermal reactor (Supplementary Note 4). Our solar thermal catalytic system maintained its initial activity and similar product distribution over at least five repetitions without replacing the catalyst and the solvent (Supplementary Fig. 32), constantly achieving complete LDPE conversion. Benefiting from the strong metal–support interaction between Cu NPs and 2D Si, the spent catalyst retained its original morphology without discernible aggregation of metallic NPs (Supplementary Fig. 32). Furthermore, the Cu content remained at 3.5% after the reaction, suggesting that the encapsulation of Cu NPs in stacked 2D Si prevented the loss of metallic species. No obvious difference can be identified for the Cu/2D Si catalyst structure after repeated reaction (Supplementary Fig. 34). These results show the potential of our solar thermal system in the long-term industrial run.

Upcycling of real-world post-consumer plastic waste was elaborately performed to explicate the applicability, potential limitations and corresponding practical solutions of this ambient solar thermal catalytic system. As most real-world plastic wastes are multicomponent mixtures, the feedstocks investigated in our work included the six most commonly used plastics—PET, high-density polyethylene (HDPE), polyvinyl chloride (PVC), LDPE, polypropylene (PP) and polystyrene (PS)—and combinations thereof.

The system demonstrated effective conversions for various real-world post-consumer polyolefins (LDPE, HDPE, PP). Low-density polyethylene from a food packaging yielded 81.3% hydrocarbons, which is only 10% lower than that of virgin LDPE powder (Supplementary Table 4). The PP surgical mask decomposition proceeded successfully and generated >70% yield of hydrocarbons, and the slightly reduced yield is attributed to the presence of non-degradable ingredients (such as carbon fibre). Upcycling of HDPE bottle, PP tube and PE/PP caps showed lower activities due to their compact polymer chains and additives introduced during their manufacture, which decreased the contact surface between catalyst and plastics. To demonstrate it, we pretreated the wastes by ball-milling them into powder, thereby increasing the catalyst-plastic contact surface. As shown in Supplementary Table 4 and Supplementary Fig. 35, this pre-treatment enhanced both the conversion and yields to varying extents. However, complete conversion was not achieved, and extending the reaction time did not result in obvious further increases. The GPC analysis (Supplementary Fig. 36) illustrated that the solid residues are not macromolecules. This suggests that residues primarily consist of additives—such as carbon fibres, colourants, plasticizers, stabilizers and flame retardants—that are not convertible within the upcycling system. Thus, residual solids, including unconverted residues and catalysts, may accumulate in the system over time and require periodic removal. Unlike most past studies using noble metals, our upcycling process employs naturally abundant, low-cost copper and silicon, making the disposal eco-friendlier. Besides, this issue could be ameliorated by first dissolving the cut plastics in boiling xylenes and subsequently precipitating them with acetone to eliminate additives and impurities[42], as demonstrated in Supplementary Fig. 37.

We then further investigated the other three kinds of plastics with the structures and compositions distinct from polyolefins. The PVC plastics contain Cl but hold the same C–C bond polymeric skeleton and similarly sized surface groups. As shown in Supplementary Figs. 38 and 39 and Supplementary Table 4, the decomposition of the PVC glove proceeded successfully in our system, releasing HCl and yielding hydrocarbons as the upcycled products. By contrast, the same approach exhibited substantially lower upcycling performance for PS, which is attributed to the phenyl groups along the polymeric skeleton that hinder the isomerization into stable tertiary carbenium ions, thus inhibiting the β-scission of the polymer chain. The unreacted PS bottle pieces remained large in size (>5 mm), which can be easily picked out from the mixture, and no substantial morphological or structural changes were observed in the Cu/2D Si catalyst after the reaction (Supplementary Fig. 40). Finally, when the PET pieces were fed in the system, the pieces fleetly disaggregate, and many white powders appeared in the system with a weight of about 2.5 times that of the input PET, and no gas or liquid products were detected. The characteristic peaks of PET were identified in the infrared spectrum of the powder (Supplementary Fig. 41). Elemental analysis revealed the presence of C (51.08 wt%), O (21.12 wt%), N (3.12 wt%), H (2.46 wt%), Si (0.47 wt%), Cu (0.03 wt%), Al (13.63 wt%) and Cl (6.91 wt%) in the powder. This phenomenon is probably due to the complexation of $[Al_2Cl_7]^-$ anions in our chloroaluminate ionic liquids with C=O bonds in PET, leading to solid product formation.

The effects of PVC, PS and PET as impurities on upcycling the mixed polyolefin plastics were also revealed. As there is already plenty of chlorine in our chloroaluminate ionic liquids, the chlorine released from PVC did not interfere with our arranged reaction (Supplementary Fig. 42 and Supplementary Table 4). The conversion of PVC-containing mixed polyolefins proceeded successfully. Furthermore, our results show that the presence of inert PS does not affect the conversion of other mixed plastics. Regarding PET, the overwhelming PET contamination suppressed the mixed plastic conversion, leading to lower product yields, which is probably attributed to the detriment of PET to the ionic liquid described earlier; however, minor contamination did not affect the hydrocarbon yields.

Considering the potential fouling effect of PET, we hereby propose a facile and scalable slicing-separation-conversion protocol to achieve a total solution for upcycling polyolefins from real-world mixed wastes. We first separated LDPE, HDPE and PP plastics from other heavier-than-water polymers such as PET, PVC and PS, using a simple sink-float method in water, simultaneously removing surface dust (Supplementary Fig. 43). The waste plastics were then cut into small pieces and fed into the reactor. The conversion of mixed post-consumer polyolefins proceeded successfully under mild conditions, producing hydrocarbons as the upcycled product (Fig. 5a). These results confirmed our protocol's compatibility with post-consumer products and proved our system robust enough to yield substantial hydrocarbons in the presence of additives.

Following the successful laboratory practice, we performed a larger-scale outdoor demonstration under natural sunlight to verify the feasibility and scalability of the solar thermal catalytic polyethylene upcycling. Five grams of chopped LDPE bag pieces (M_w = 270,598 g mol^{-1}; Supplementary Fig. 44), 30 mmol of $[C_4Py]Cl$, 60 mmol of $AlCl_3$, 15 ml of chloroform, 50 µl TBC initiator and 50 mg of Cu/2D Si were loaded into a 300 ml reactor. As shown in Fig. 5b, sunlight was concentrated by a poly(methyl methacrylate) Fresnel lens and then passed through a quartz window on the upper side of the reactor. By adjusting the distance between the reactor and the Fresnel lens, the incident light intensity was controlled between 4 to 5 sun (400 to 500 mW cm^{-2}), driving the reactor temperature up to 55 °C. After 6 h of reaction, the LDPE pieces completely disappeared and yielded 85.8% of hydrocarbons with this feasible set-up, omitting any expensive solar concentrators and heating equipment.

A possible concern for the application of the upcycling system is the product separation from the reactant–catalyst mixture. However, considering the extremely high boiling point of the ionic liquid, most of the hydrocarbons can be separated from the solvent via a facile and scalable distillation method[31]. Besides, past reports have demonstrated that the conjunct unsaturated hydrocarbons can be easily converted into saturated hydrocarbons via hydrogenation, allowing the extraction of remanent product[13,43].

To assess the potential of our strategy, we performed a techno-economic analysis and LCA of a conceptual solar thermal upcycling facility capable of processing 30,000 tonnes per year of sorted LDPE waste. A detailed description and conceptualized process are given in Supplementary Note 5 and Supplementary Fig. 45. The plastics are converted by the above-mentioned outdoor solar thermal catalytic system, and the product distribution is presented in Supplementary Fig. 46. The resulting gas product—which comprises propane, *n*-butane and iso-butane—can be easily separated through condensation and sold (because we hardly generate other light hydrocarbons, including methane and ethane). As for the unsaturated cyclic hydrocarbons, they can be separated from the ionic liquid with a facile hydrogenation step[43], and gases, naphtha, diesel and paraffin wax are separated using a distillation column, whereas the ionic liquid solvent is recovered. Considering that the prices of plastic waste and fuels vary considerably across regions, we collected pricing data from two representative markets: China, the world's largest plastic producer, and the United States, the world's largest oil producer. With a high yield towards valuable hydrocarbons, the product revenues could reach US$678.19 and US$533.94 per tonne of LDPE in China and the United States, respectively. The minimum selling price (MSP) of our primary product, naphtha and

cost breakdown with variable LDPE waste prices are shown in Fig. 5c,d (see Supplementary Tables 5 and 6 for details). Due to the smaller scale of plastic upcycling plants compared with petrochemical facilities, the amortization and fixed costs (operating labour, supervision and overhead) hold the largest portion of costs. Notably, our mild operating conditions enable a minuscule total utilities. As the world's largest plastic producer and consumer, China presents a low LDPE waste price range from US$83.1 per tonne to US$150 per tonne. In the China case, the MSP of naphtha is estimated at US$0.54–0.71 per kg, which is much lower than the domestic market price of petroleum-derived naphtha ($1.05 per kg). In the United States, plastic waste collection costs range from $100–300 per tonne[44], resulting in a naphtha MSP of US$0.37–0.89 per kg, whereas America's domestic market price of petroleum-derived naphtha is US$0.65 per kg. This suggests that the solar thermal recycling plant features economically viable only when plastic waste prices remain below US$210.07 per tonne in the United States. Overall, our solar thermal LDPE upcycling route shows profitable promise in regions with higher energy prices or lower waste costs, while improved waste collection strategies or government subsidies may be necessary to improve economic feasibility in certain regions.

A comparative LCA analysis was conducted to illustrate the environmental impact of solar thermal upcycling for fuel production from plastic waste on a cradle-to-grave scope. The results of the global warming potential (GWP) are graphically shown in Fig. 5d, with more extensive analysis developed in Supplementary Figs. 47 and 48 and Supplementary Tables 9–11. The solar thermal LDPE upcycling route demonstrates positive environmental effects across various categories, including abiotic fossil fuel depletion, marine aquatic ecotoxicity, ozone layer depletion, global warming, acidification and eutrophication. It is noteworthy that all of products from our route exhibit lower GWP values than those of conventional fossil-derived route, leading to a 353.5 kg of CO_2-eq emissions reduction per tonne of LDPE converted. Thus, by executing the recycling of ubiquitous polyolefin waste, a potential 10.6 kt of greenhouse gas reduction could be realized per year in our envisioned plant, contributing to the goal of carbon neutrality.

Conclusion

In summary we have demonstrated an ambient solar thermal catalytic system for selective upcycling of mixed plastic waste—including LDPE powder, LDPE bags, PP surgical masks, HDPE bottles and PVC gloves—that features mild operating conditions, high recycling efficiency and bypasses the need for co-reactants and heating equipment. The 2D Si-encapsulated Cu NPs catalyst can efficiently capture the entire solar spectrum, allowing relatively low light input of 4 sun. The diluted chloroaluminate ionic liquid was applied as the solvent to stabilize the carbo-cation intermediates. Benefiting from the photothermal local heating effect, the depolymerization of polyolefin can be efficiently performed at a relatively low temperature of 55 °C without external heating. The polyethylene can be completely transformed into valuable hydrocarbons with a total yield of 91%. Notably, the product consists of separable alkanes (C_3–C_7) and cyclic hydrocarbons (C_8–C_{26}), whereas low-value methane and ethane are undetected. The selective plastic cracking is attributed to a combination of C–C bond β-scissions and intramolecular cyclizations. The Cu NPs serve as the active sites for dehydrogenation process and 2D Si promote light absorption, which concertedly enhanced the conversion efficiency. Following a facile and scalable protocol, our approach also achieved high-performance upcycling of real-world polyolefin post-consumer products either alone or in a mixed form, containing impurities such as additives, labels and even other common plastics. Moreover, the solar thermal catalytic upcycling system demonstrated robustness through repeated experiments and good scalability through outdoor tests. It also demonstrates a pronounced energy advantage over the thermal route via renewable electrification. The techno-economic analysis and LCA results further demonstrated its economic viability and eco-friendly nature, indicating potential for industrial implementation.

Methods

Chemicals and materials

All of the chemicals were used as received without additional purification. *N*-butyl pyridinium chloride ([C_4Py]Cl, 98%) and calcium silicide ($CaSi_2$, 99%) were purchased from Macklin. Copper nitrate trihydrate (99.99%) and TBC were purchased from Aladdin. Hydrochloric acid (37 wt%), anhydrous aluminium trichloride ($AlCl_3$), trichloromethane ($CHCl_3$) and sodium hydroxide (NaOH) are analytical reagents that were purchased from Sinopharm Chemical Reagent. Low-density polyethylene powder was purchased from Dongguan Jinheng Plastic.

Preparation of 2D Si

Two-dimensional Si was exfoliated from bulk calcium silicide according to our previously reported method[21]. Hydrochloric acid (400 ml) was added into a jacketed flask and purged with nitrogen for 30 min to eliminate oxygen. The reaction was kept at between −20 °C and −25 °C by circulating liquid (ethylene glycol/water) in the interlayer of the flask. Calcium silicide powder (4.00 g) was then added to hydrochloric acid and stirred under a nitrogen environment for one week. The remaining solid (stacked 2D Si) was separated by centrifugation and washed thoroughly with ethanol and vacuum-dried at 60 °C.

Preparation of 2D Si-encapsulated Cu NPs

The Cu NPs were encapsulated by the 2D Si via a modified two-step impregnation-annealing method. Two-dimensional Si (0.50 g) was dispersed in an ethanol solution containing 0.86 g of $CuCl_2{\cdot}2H_2O$ (6 wt% metal feeding). The mixture was sonicated for 30 min and then maintained at 80 °C to remove solvent. The obtained solid was annealed at 300 °C with a temperature ramp of 5 °C min^{-1} for 5 h in a nitrogen environment, allowing the bonding of metal precursors to the 2D Si support. The powder was then washed with 30 ml ethanol and centrifugation-collected three times to remove unbound metal precursors. Finally, after drying in an oven, the resulting solid was annealed at 550 °C at a heating rate of 2 °C min^{-1} for 5 h in nitrogen flow.

Procedure for solar thermal catalytic plastic upcycling

The highly polar ionic solvent was synthesized according to a reported method. In a typical preparation of highly polar reaction solvent, 6 mmol of *N*-butyl pyridinium chloride ([C_4Py]Cl), 12 mmol of anhydrous $AlCl_3$ and 3 ml of trichloromethane were mixed in a quartz flask. Note that the preparation process was performed in a nitrogen atmosphere and cooled with an ice-water bath. A homogeneous light-grey liquid was obtained under magnetic stirring. Then, 10 mg of Cu/2D Si, 200 mg of LDPE, and 10 µl of TBC were dispersed in the solvent. The TBC acted as an initiator to offer an initial concentration of carbo-cations. The reactor was then placed under a simulated sunlight illumination from a xenon lamp (PLS-SXE 300D or PLS-SME300E H1, Beijing Perfectlight Technology) equipped with a AM 1.5 optical filter with a light intensity of 4 sun, stimulating the photothermal effect of Cu/2D Si to reach the specific reaction temperature. The temperature was kept for a 6 h reaction time.

Procedure for catalytic plastic upcycling under other conditions

In the thermal catalytic upcycling of LDPE, the incident illumination was removed and the reactor was maintained at certain temperature using external heating. In the UV + thermal catalytic upcycling of LDPE, the reaction conditions were the same with those of thermal catalysis, but with extra UV light with an intensity of 251 mW cm^{-2} generated from a mercury lamp. In the robustness experiments, after a 6 h of solar thermal plastic upcycling reaction over Cu/2D Si, the gaseous product in the headspace was collected and analysed by gas chromatograph.

The reactor was then refilled with 200 mg of LDPE powder in each repeated run. Note that the ionic liquid was quenched in the quantitative determination of liquid product. Therefore, only one sampling for liquid product analysis was tested at the end of the several repetitions.

In the upcycling of real-world plastics, the commercial LDPE bag, PP surgical mask, HDPE bottle, PVC glove and PS foam were cut into small pieces of about 0.5 cm for further experiments. In the reaction, the LDPE powder was substituted by 200 mg other plastics and with a 6 h reaction time for the LDPE bag, PP surgical mask, PVC glove and PS foam, and a 12 h reaction time for HDPE bottle.

In the outdoor experiment, 30 mmol of *N*-butyl pyridinium chloride ($[C_4Py]Cl$), 60 mmol of anhydrous $AlCl_3$ and 15 ml of trichloromethane were added to a 300 ml reactor. The mixture was transformed to a light-grey liquid under magnetic stirring in a nitrogen atmosphere and an ice-water bath. Then, 50 mg of Cu/2D Si, 5 g of chopped LDPE bag pieces and 50 µl of TBC were dispersed in the solvent. The sunlight concentrated by a poly(methyl methacrylate) Fresnel lens passed through a quartz window on the upper side of the reactor, and the incident light intensity was controlled between 4 to 5 sun. The illumination was kept for a 6 h reaction time.

Procedure for upcycling product analysis

For the quantitation of gaseous product in the headspace, the reactor was connected to a gas chromatograph (Inesa, GC-128). Gaseous C_3 to C_7 hydrocarbons were detected and analysed using an Al_2O_3 capillary column with a flame ionization detector.

For the quantitation of liquid product in the solvent, the liquid product was extracted with chloroform and analysed using a gas chromatograph (Inesa, GC-128) equipped with a Supel PLOT-fused silica capillary column (30 m × 0.53 mm × 2.0 µm), which was calibrated by standard hydrocarbons (C_5–C_{30} normal alkanes). An extracting agent (15 ml chloroform) and external standard (10 µl mesitylene) were injected into the reactor. To thoroughly extract the liquid product, the tube reactor was cooled to −15 °C in a refrigerator, and injected with 10 ml of 4 M NaOH aqueous solution to quench the ionic liquid (the reactor was maintained in refrigerator to keep the low temperature, minimizing the volatilization of liquid product). The remaining polymeric solid swelled in the chloroform phase, and was collected by filtration. The liquid-phase products were injected into a vaporization chamber using microsyringe and analysed by gas chromatograph. It is virtually impossible to figure out every peak in the chromatogram due to the lack of calibrations of every kind of multifarious isomers. According to reported methods[45], the chromatographic peaks with retention times between that of normal C_{n-1} and C_n were assigned to signals of C_n. It was assumed that alkane isomers shared the same calibration factor as their corresponding normal alkanes. The conversion of plastics was evaluated gravimetrically as following:

$$\text{Conversion}\,(\%) = \frac{(m(\text{plastic}) - m(\text{remaining polymeric solid}))}{m(\text{plastic})} \times 100\% \quad (1)$$

where m(plastic) is the mass of the initial plastic feedstocks, and m(remaining polymeric solid) is the mass of the remaining polymeric solid swelled in the chloroform phase after reaction.

The yields of products were calculated on the basis of the carbon element recovery efficiency as following:

$$\text{Product yield}\,(\%) = \frac{n\,(\text{carbon in } C_n \text{ product})}{n\,(\text{carbon in plastic})} \times 100\% \quad (2)$$

where n(carbon in plastic) represents the moles of carbon elements in the initial plastic feedstocks, and n(carbon in C_n product) is the mol of carbon elements in C_n product. The yields of alkanes and unsaturated hydrocarbons are the sum of the yields of the corresponding C_n product classes.

Furthermore, depending on the molecular formula calculated from GC–MS patterns, the products were classified into four categories: cycloalkanes, cycloolefins, aromatics and conjunct hydrocarbons (the compounds with more than four degrees of unsaturation, presenting a structure integrated from the first four categories). The selectivity of C_{nx} products (that is, the products with n carbon number and matching to formula of x category) was calculated as following:

$$\text{Product selectivity}\,(\%) = \frac{n\,(\text{carbon in} C_{nx} \text{ product})}{n\,(\text{carbon in product})} \times 100\% \quad (3)$$

where n(carbon in product) is the moles of carbon elements in total products, and n(carbon in C_{nx} product) is the mol of carbon elements in C_{nx} product. The selectivities of certain category are the sum of the selectivities of the corresponding C_{nx} product classes.

Characterization

Transmission electron microscopy images and EDX mappings were obtained with an FEI-Tecnai F20 (200 kV) equipment. The XAS spectra at the Cu K-edge of catalyst and references were collected at Beijing SCISTAR. The metal content in various samples was determined by inductively coupled plasma mass spectrometry (Aurora M90, Jenoptik). The XRD pattern was measured on an XRD-700 X-ray diffractometer from Shimadzu. XPS were obtained using a Thermo Scientific K-Alpha. Fourier transform infrared experiments were conducted using Thermo Nicolet iS50. The molecular weights (M_w) and dispersities (Đ) of the plastics were determined by an Agilent 1260 Infinity II high-temperature size-exclusion chromatography system and two PLgel Mixed-B columns (with M_w values between 500 and 10,000,000). 1,2,4-Trichlorobenzene containing 0.0125% (w/w) 2,6-di-*tert*-butyl-4-methylphenol was employed as an eluent at a flow rate of 1 ml min^{-1} at 150 °C. Commercial polystyrenes were used as the calibration standards.

Heat transfer stimulation

The commercial software COMSOL Multiphysics[46] was applied to simulate the heat transfer of Cu/2D Si catalyst immersed in $CHCl_3$. In the simulation, a 2D Si nanosheet is set to a rectangular piece of 400 nm × 400 nm × 30 nm, and 16 Cu particles with a radius of 16 nm are equidistantly dispersed on its surface. The catalyst particulate is simplified as a sandwich structure of five layers of silicon nanosheet encapsulating five layers of Cu particles. The entire structure is placed in a 2,000 nm × 2,000 nm × 2,000 nm cuboid filled with $CHCl_3$ to simulate the heat transfer under $CHCl_3$ immersion.

Data availability

The datasets generated during and/or analysed during the study are provided with this paper or available from the authors on reasonable request. Source Data are provided with this paper.

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Author contributions

C.X., S.W., W.S. and G.A.O conceived and designed the experiments. C.X., L. Wu and S.W. peformed the synthesis experiments. C.X., D.Z. and D.K. performed the structural characterization of catalysts. C.X., W.S. performed and analysed the catalytic experiments. With the support of L. Wang, C.X. performed the hydrogenation experiments. Y.Z. and C.L. performed the heat transfer stimulation. C.X., C.M., Di Yang, W.G.

and W.W. conducted the techno-economic analysis. C.X., W.S., C.M., Deren Yang and G.A.O. wrote the paper. W.S., G.A.O and Deren Yang supervised the project. All authors commented on the final paper.

Competing interests

A Chinese patent application (application no. 202410188427.2; Inventors: W.S., C.X., Deren.Y.) and a United States patent application (application no. 19057203; Inventors: W.S., C.X., G.A.O., Deren.Y.) related to part of this work are currently under examination. The other authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Geoffrey A. Ozin, Deren Yang or Wei Sun.

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[1]State Key Laboratory of Silicon and Advanced Semiconductor Materials, School of Materials Science and Engineering, Zhejiang University, Hangzhou, China. [2]Materials Chemistry and Nanochemistry Research Group, Solar Fuels Cluster, Departments of Chemistry, University of Toronto, Toronto, Ontario, Canada. [3]State Key Laboratory of Green Papermaking and Resource Recycling, School of Environmental Science and Engineering, Shanghai Jiao Tong University, Shanghai, China. [4]Institute of Functional Nano and Soft Materials (FUNSOM), Jiangsu Key Laboratory for Carbon-Based Functional Materials & Devices, Soochow University, Suzhou, China. [5]Key Lab of Biomass Chemical Engineering of Ministry of Education, College of Chemical and Biological Engineering, Zhejiang University, Hangzhou, China. [6]Institute of Advanced Semiconductors & Zhejiang Provincial Key Laboratory of Power Semiconductor Materials and Devices, Hangzhou Global Scientific and Technological Innovation Center, Hangzhou, China. [7]Shangyu Institute of Semiconductor Materials, Shaoxing, China. [8]These authors contributed equally: Chuanwang Xing, Chengliang Mao. ✉e-mail: g.ozin@utoronto.ca; mseyang@zju.edu.cn; sunnyway423@zju.edu.cn